

Disaster Management: The Role *of* FOSS



Climate change is gathering momentum and because of this, the world has witnessed some of the most devastating disasters recently. Tsunamis, cyclones, floods, earthquakes and even wildfires have destroyed lives and entire regions, of late. So can we look to open source software to help mitigate the effects of these disasters?

A disaster is a sudden accident or catastrophe, like an earthquake, a flood or a fire, which the community is incapable of coping with. It could be a natural or a man-made event, which kills people and destroys goods, services and the environment. In such circumstances, help from the government and information agencies is required.

In disaster risk management (DRM), financing and insurance are two major concerns for the government.

Natural disasters cause damage to human life and property. They result in severe losses. Some recent examples include the Kerala floods, the Northern California fires, hurricanes Irma and Harvey, and the volcanic eruption in Hawaii.

In a disaster situation, aid institutions must keep track of the victims' needs and respond accordingly. While monitoring large scale disasters, massive amounts of heterogeneous data gets collected and that must be managed efficiently by aid agencies. During a disaster, if information is not handled well, help can get delayed or become ineffective, leading to the potential waste of resources or in uneven aid distribution. FOSS can help manage disasters, and has the potential to improve the efficiency and effectiveness of aid agencies.

During disasters, effective information systems are required because they can provide timely access to comprehensive, relevant and reliable solutions to victims. FOSS for disaster

management is in high demand because of its lower cost and higher efficiency.

FOSS in disaster management

The Asia-Pacific region, with 60 per cent of the world's population, is highly vulnerable to disasters. Between 2002 and 2011, 40 per cent of all reported disasters (1,524) occurred in Asia, causing 63 per cent of all global disaster deaths (669,263), and causing 48 per cent of global damage (costing US\$ 691 billion).

FOSS helps the disaster risk management (DRM) team in the following ways:

- Protecting lives
- Protecting property (buildings, infrastructure, etc)
- Protecting livelihoods at the